# Benchmarking Multi-Step Sensor Forecasting for Threshold-Based Predictive Maintenance: An Integrated IoT-to-Enterprise Framework

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***Abstract*—Sensor forecasting can support maintenance planning when forecasts are connected to monitoring, threshold alerts, and service workflows. This paper presents an integrated IoT-to-enterprise prototype and a comparative evaluation of 24-hour multi-step sensor forecasting. The prototype combines ESP32-S3 acquisition, GPS, DS18B20 temperature sensing, BLE tyre-pressure monitoring, HTTP telemetry, and the Swift platform, with Salesforce alert ingestion and status feedback. Its documented maintenance workflow uses Cases and Flow; the available repository evidence supports integration components rather than a measured operational deployment. A corrected benchmark evaluates persistence, seasonal naive, Random Forest, and a bidirectional long short-term memory (Bi-LSTM) network on three synthetic sensor datasets, using seven days of historical target values and chronological partitions with disjoint target periods. We add five-seed Bi-LSTM robustness, an LSTM/Bi-LSTM architecture comparison, causal feature ablation, proxy threshold-event scoring, moving-block bootstrap intervals, and external sensor-forecasting evaluation on the NASA C-MAPSS FD001 simulator benchmark. Random Forest achieves the lowest mean absolute error for engine temperature (1.708°C) and tyre temperature (2.897°C), while persistence performs best for tyre pressure (0.558 PSI). The C-MAPSS and feature experiments use a separately recorded TensorFlow/scikit-learn environment, which is retained as a reproducibility limitation. The evaluation does not establish failure prediction, real-fleet deployment, or maintenance-outcome benefits.**

***Index Terms*—Internet of Things, multi-step forecasting, predictive maintenance, Random Forest, Bi-LSTM, enterprise maintenance workflow.**

## I. Introduction

Vehicle maintenance decisions require both visibility into asset condition and a process for acting on that information. A dashboard can display abnormal readings, but an actionable maintenance workflow also needs an asset identifier, an alert description, an assigned service task, and feedback on its status. Predictive-maintenance architectures therefore combine data acquisition, analysis, and decision processes; forecasting quality is only one part of this chain [1].

The intended application of AutoSense is vehicle monitoring, including the aviation ground-support context that motivated the prototype. The available evidence does not establish deployment in an operational aviation fleet. We study the narrower task of forecasting engine temperature, tyre pressure, and tyre temperature over the next 24 hours, with forecasts intended to inform threshold-based maintenance alerts. Predicting a threshold excursion is distinct from predicting a component failure: no failure-time, remaining-useful-life, or maintenance-outcome labels are available here.

The forecasting study compares models under a shared protocol instead of assuming that a recurrent neural network is the strongest choice. Straightforward forecasting methods provide essential reference points [2]. We separate two kinds of evidence: repository artifacts describing a telemetry-to-service prototype, and numerical results from a reproducible synthetic-data benchmark. This separation prevents interface demonstrations from being interpreted as forecast validation.

The contributions are: (1) an IoT-to-maintenance workflow prototype connecting telemetry, monitoring, forecasting interfaces, alerts, and enterprise status feedback; (2) a standardized 24-hour forecasting benchmark for three sensor variables with consistent information access and disjoint target periods; (3) robustness, architecture, feature, threshold-event, and dependence-aware uncertainty analyses; (4) a C-MAPSS FD001 external simulated-benchmark sensor-forecasting study with engine-level separation; and (5) an evidence-bounded account of Salesforce maintenance integration. The contribution is comparative evaluation and system integration, rather than a new forecasting algorithm or a claim of Bi-LSTM superiority.

## II. Related Work

### A. Predictive Maintenance and Fleet Monitoring

Condition-based maintenance is commonly framed as data acquisition, data processing, and maintenance decision-making [3]. Reviews of machine-learning predictive maintenance likewise emphasize that performance is task- and data-dependent rather than a property of a model family in isolation [1], [4]. This motivates separating the present forecasting metrics from the unmeasured downstream maintenance decision. Fleet monitoring is the application context here; labelled maintenance outcomes would be required to evaluate breakdown avoidance, cost, or downtime.

### B. IoT Maintenance Architectures

Cloud-enhanced maintenance architectures connect acquisition and analytical components [1]. In our implementation, an

ESP32-S3 provides the acquisition and wireless communication platform, while the DS18B20 provides a digital temperature interface [5], [6]. These component references establish device capabilities, not automotive qualification, sensing accuracy in the installed configuration, or end-to-end reliability. The prototype's concrete evidence consists of firmware, a circuit schematic, interface captures, and integration code.

### C. Sensor Time-Series Forecasting

Naive and seasonal naive methods establish useful forecasting benchmarks [2]. Random Forest combines randomized tree predictors [7], providing a nonlinear model of lagged target values without recurrent state. LSTM introduces gated recurrent memory [8], while bidirectional recurrent processing encodes a sequence in both temporal directions [9]. In forecasting, both directions may process the observed input window without accessing future targets. These methodological foundations justify the comparison; they do not imply superior predictive-maintenance performance for a particular architecture.

### D. External Simulated Degradation Benchmarks

NASA's C-MAPSS repository provides simulator-generated run-to-failure turbofan trajectories under controlled operating and fault settings [10]. Many C-MAPSS studies target remaining useful life (RUL). That task is deliberately different from ours: we forecast four observed sensors ten cycles ahead from complete FD001 training trajectories, and use normalized life only after prediction to stratify error. Thus no C-MAPSS RUL score or physical-flight-telemetry claim is made.

### E. Enterprise Maintenance Workflow Integration

Salesforce Flow supports record-triggered criteria and record-creation actions [11]. Such mechanisms can translate an alert into a service workflow. Product documentation establishes that capability, whereas deployment-specific validation requires configuration exports and execution evidence. We describe the available alert-ingestion and vehicle-status code separately from the Case/Flow linkage documented in the original project narrative.

## III. System Architecture

Fig. 1 summarizes the logical integration. The on-vehicle sensor layer combines GPS location, DS18B20 temperature sensing, and BLE TPMS pressure and temperature readings. The ESP32-S3 firmware contains GPS parsing, OneWire temperature acquisition, BLE advertisement parsing, JSON payload construction, and HTTP POST transmission. The stored PCB schematic documents the hardware design. Neither its existence nor the firmware demonstrates environmental qualification or calibrated engine-condition measurement.

The Swift platform is the telemetry and visualization layer. Repository scripts fetch vehicle and sensor records, construct forecasting inputs, and format predictions for platform ingestion. Those integration scripts reference legacy neural models. The corrected offline benchmark reported here is a separate analysis; it has not been substituted into the deployed prototype or validated through a live forecast endpoint. In particular, the benchmark's time alignment must not be inferred from legacy dashboard timestamps.

Threshold logic connects readings or forecast excursions to alerts. For a target with configured acceptable bounds $[a, b]$, a possible forecast-alert rule is

$$A_t = \mathbb{1}\{\exists h \in \{1, \ldots, H\} : \hat{y}_{t+h|t} < a \text{ or } \hat{y}_{t+h|t} > b\}. \quad (1)$$

Equation 1 states the intended threshold semantics. Bounds must be appropriate to the sensor placement and asset; a threshold displayed in a screenshot is a configured prototype value, not a validated safety limit. We do report offline precision, recall, F1, false-positive rate, and an origin-level lead-time convention for training-derived proxy excursion labels. Those labels are neither failures nor certified engineering alarms, and no maintenance effectiveness is estimated.

Salesforce is the maintenance/service workflow layer. The available Apex code imports alerts into a custom Vehicle Alert object, checks existing alert identifiers before inserting records, and sends vehicle-status updates back to Swift through an asynchronous callout. The original project describes Cases and Flow for maintenance-task creation and resolution feedback. Because those Flow definitions are not exported in the repository, that part of the diagram remains a documented integration design, rather than independently reproduced Case automation.

## IV. Forecasting Methodology

### A. Synthetic Data and Time Validation

All three benchmark series are synthetic. The repository's generators reproduce their timestamps exactly and their target values to floating-point precision (maximum absolute discrepancy below $1.5 \times 10^{-14}$). Thus, the presence of physical sensor hardware does not establish sensor-collected provenance for these datasets. Each file covers 2024 from January 1 through December 31 at 00:00 in its recorded clock; the final day is not fully sampled. No timezone is inferred.

The corrected protocol rejects missing timestamps, duplicate or unsorted timestamps, non-finite target values, and irregular sampling. No implicit sorting, interpolation, or resampling is performed. Engine temperature is hourly; tyre pressure and tyre temperature are sampled every three hours. Table II records the resulting window counts. A seven-day input contains 168 hourly samples or 56 three-hour samples. A 24-hour output contains 24 hourly steps or eight three-hour steps. Horizon labels denote time after the last observed input sample.

### B. Chronological Partitioning

For a series of $N$ raw rows, the test region starts at $b_t = \lfloor 0.8N \rfloor$ and validation starts at $b_v = \lfloor 0.85 b_t \rfloor$. These boundaries allocate approximately 68% to fitting, 12% to validation, and 20% to testing. A window is assigned only if its entire target sequence lies inside one region. Windows with

TABLE I
COMPACT, VERIFIABLE POSITIONING OF THIS STUDY. "DECISION/WORKFLOW" MEANS AN EVALUATED DOWNSTREAM DECISION, NOT MERELY AN ARCHITECTURE DIAGRAM.

| Study | Setting and provenance | Forecasting or maintenance scope | Methods and metrics | Decision/workflow evidence |
|---|---|---|---|---|
| Jardine *et al.* [3] | Mechanical systems; review literature | Condition monitoring, diagnostics, and prognostics | Review of models and algorithms; no single metric set | Reviews maintenance decision support |
| Carvalho *et al.* [4] | Industrial PdM; systematic-review literature | Machine-learning predictive maintenance | Review of ML methods; study-dependent metrics | No common workflow experiment |
| Saxena and Goebel [10] | Simulated turbofan engines; NASA C-MAPSS | Run-to-failure benchmark resource | Dataset resource, not a model study | None |
| This work | Synthetic AutoSense; simulated C-MAPSS FD001 | Multi-step sensor forecasting; proxy threshold events | Naive, RF, LSTM, Bi-LSTM; MAE, RMSE, $R^2$, event metrics | Prototype artifacts only; no evaluated maintenance outcome |

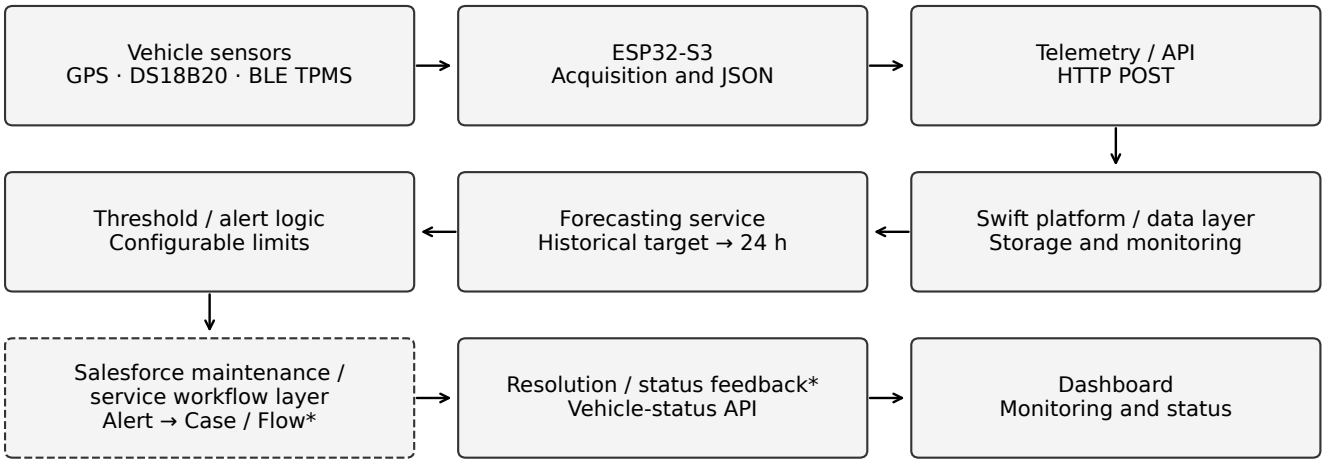



Fig. 1. Logical IoT-to-enterprise architecture. Arrows represent intended information flow rather than measured transaction traces. *Case/Flow automation and its resolution linkage are documented in the original project narrative; exported Flow configuration and execution logs are unavailable. Alert ingestion and vehicle-status feedback have repository code.

targets crossing either boundary are excluded. Consequently, training, validation, and test target timestamps do not overlap.

Inputs may include observations from an earlier region, since they are historical information available at that forecast origin. Test origins advance one sample at a time, and later test inputs may contain previously observed test-period values. This is rolling-origin evaluation with fixed trained parameters, not a single open-loop forecast from the start of the holdout. There is no refitting on validation or test observations. Within each partition, overlapping target windows produce dependent errors.

All four models share test origins and targets. Both learned models use only the historical target sequence, with no GPS, calendar, cross-sensor, or engineered covariates. The Random Forest uses the sequence as a lag vector; the network uses it as a univariate sequence. Min–max scaling for the network is fitted only on the fitting-region raw values. Forecasts are inverse-transformed before scoring. This protocol also avoids the earlier baseline utility's target overlap caused by splitting already-constructed windows. Legacy notebook results with different features or splits are not mixed into the comparison.

### C. Forecasting Models

Persistence repeats the latest observed target across the horizon. Seasonal naive repeats the immediately preceding 24-hour pattern; the seasonal period is therefore 24 samples for engine temperature and eight for the tyre targets. Both baselines require no parameter fitting.

Random Forest is a direct multi-output regressor with 300 trees, maximum depth 16, minimum leaf size two, and random state 42. It predicts the whole output vector from the lag vector

TABLE II
DATASET AND PROTOCOL SUMMARY. TRAIN, VALIDATION, AND TEST COLUMNS COUNT COMPLETE FORECAST WINDOWS; EVERY DATASET IS SYNTHETIC.

| Target | Type | Rows | Step (h) | Lookback (h) | Horizon (h) | Train | Val. | Test | Unit |
|---|---|---|---|---|---|---|---|---|---|
| Engine temperature | Synthetic | 8761 | 1 | 168 | 24 | 5765 | 1029 | 1730 | °C |
| Tyre pressure | Synthetic | 2921 | 3 | 168 | 24 | 1922 | 344 | 578 | PSI |
| Tyre temperature | Synthetic | 2921 | 3 | 168 | 24 | 1922 | 344 | 578 | °C |

and uses the same fitting windows as the neural model. The unchanged implementation uses scikit-learn defaults for other parameters.

The evaluated network, labelled Bi-LSTM, contains a bidirectional LSTM with 128 units per direction and sequence output, batch normalization, dropout 0.2, a second bidirectional LSTM with 64 units per direction and sequence output, batch normalization, dropout 0.3, a unidirectional LSTM with 32 units, a dense ReLU layer with 64 units, dropout 0.2, and a linear output layer of size $H$. Bidirectionality is confined to historical inputs. The architecture is a mixed recurrent stack, not a claim that every recurrent layer is bidirectional.

### D. Evaluation Metrics

Let $M$ be the number of test origins, $H$ the number of forecast steps, and $e_{ih} = y_{ih} - \hat{y}_{ih}$. Metrics are pooled over all origin–horizon pairs:

$$\mathrm{MAE} = \frac{1}{MH} \sum_{i=1}^{M} \sum_{h=1}^{H} |e_{ih}|, \tag{2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{MH} \sum_{i=1}^{M} \sum_{h=1}^{H} e_{ih}^2}, \tag{3}$$

$$R^2 = 1 - \frac{\sum_{i,h} e_{ih}^2}{\sum_{i,h} (y_{ih} - \bar{y})^2}, \tag{4}$$

where $\bar{y}$ is the pooled target mean. MAE and RMSE retain physical units; $R^2$ is dimensionless. Negative $R^2$ means worse squared error than the pooled-mean reference. Horizon-wise MAE averages across origins at each step. Pooling counts a timestamp more than once when predicted at different horizons; it is not a metric over unique timestamps.

## V. EXPERIMENTAL SETUP

The corrected single-seed benchmark, five-seed AutoSense Bi-LSTM study, and architecture ablation record TensorFlow 2.18.0 and scikit-learn 1.9.0; the five-seed study also records NumPy 2.0.2 and pandas 2.2.2. The C-MAPSS forecasting and feature-ablation runs record TensorFlow 2.20.0 and scikit-learn 1.6.1. The C-MAPSS stage analysis records TensorFlow 2.18.0 and scikit-learn 1.9.0 but consumes forecasts from the 2.20.0/1.6.1 forecasting run. This heterogeneous environment is a documented reproducibility limitation, not a claim of uniform software execution.

Random seeds are fixed to 42 unless a phase explicitly uses the predeclared five AutoSense seeds (11, 21, 42, 84, 168) or three C-MAPSS Bi-LSTM seeds (11, 42, 84). Neural training uses Adam, mean squared error loss, batch size 64, no training-window shuffling, a 20-epoch cap, and validation-loss early stopping with patience five and restored best weights. The feature study fixes F0 (target history), F1 (target plus deterministic calendar), and F2 (target, calendar, and causal lag/rolling features); it never uses centered windows. No validation/test values select features, sensors, scalers, thresholds, or hyperparameters.

The scientific source is the committed `research_v2/results` and `research_v2/ieee_results` artifact sets. Archive validation checks source/data/protocol hashes, row counts, unique experiment keys, seed sets, and recorded environments. For C-MAPSS FD001 it additionally checks the fixed 70/15/15 engine-level split, training-only selected sensors (9, 11, 14, and 12), 30-cycle lookback, 10-cycle horizon, and that all 44 prediction NPZs contain only pre-registered test-engine IDs. Figure and table source hashes are stored with the generated assets.

## VI. RESULTS AND DISCUSSION

### A. Aggregate Comparison

Table III reports the complete corrected comparison, and Fig. 2 shows MAE with separate axes for the different target units. Random Forest achieves the lowest engine-temperature MAE (1.708°C) and RMSE (2.262°C), ahead of seasonal naive (2.623°C MAE) and Bi-LSTM (6.199°C MAE). Persistence has the largest engine MAE, 10.157°C.

For tyre pressure, persistence is strongest, with MAE 0.558 PSI and RMSE 0.979 PSI. Random Forest improves over seasonal naive in MAE but does not improve over persistence. Bi-LSTM has MAE 1.178 PSI, the largest of the four. The result shows why even a nonlinear autoregressive model needs comparison with a latest-value reference.

For tyre temperature, Random Forest again leads, with MAE 2.897°C and RMSE 3.680°C. Seasonal naive follows at 3.897°C MAE. Bi-LSTM has MAE 6.454°C and negative $R^2$, and also performs worse than persistence. Across the three targets, Bi-LSTM beats the best baseline in zero cases. These are descriptive differences for this experiment; dependent forecasts and a single run do not support a statistical-significance claim.

### B. Representative Forecast and Horizon Errors

Fig. 3 compares actual, Random Forest, and Bi-LSTM engine temperatures for one 24-hour test window. Selection is deterministic: compute Random Forest MAE for every test

TABLE III
CORRECTED TEST PERFORMANCE IN PHYSICAL UNITS. LOWER MAE/RMSE AND HIGHER $R^2$ ARE BETTER. BOLD MARKS THE LOWEST MAE AND RMSE WITHIN EACH TARGET.

| Target (unit) | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| Engine temperature (°C) | Persistence | 10.157 | 12.474 | -0.799 |
| | Seasonal naive | 2.623 | 3.449 | 0.862 |
| | Random Forest | **1.708** | **2.262** | 0.941 |
| | Bi-LSTM | 6.199 | 7.356 | 0.374 |
| Tyre pressure (PSI) | Persistence | **0.558** | **0.979** | 0.804 |
| | Seasonal naive | 0.781 | 1.241 | 0.686 |
| | Random Forest | 0.736 | 1.003 | 0.795 |
| | Bi-LSTM | 1.178 | 1.443 | 0.575 |
| Tyre temperature (°C) | Persistence | 5.786 | 7.105 | -0.840 |
| | Seasonal naive | 3.897 | 4.964 | 0.102 |
| | Random Forest | **2.897** | **3.680** | 0.507 |
| | Bi-LSTM | 6.454 | 7.862 | -1.253 |

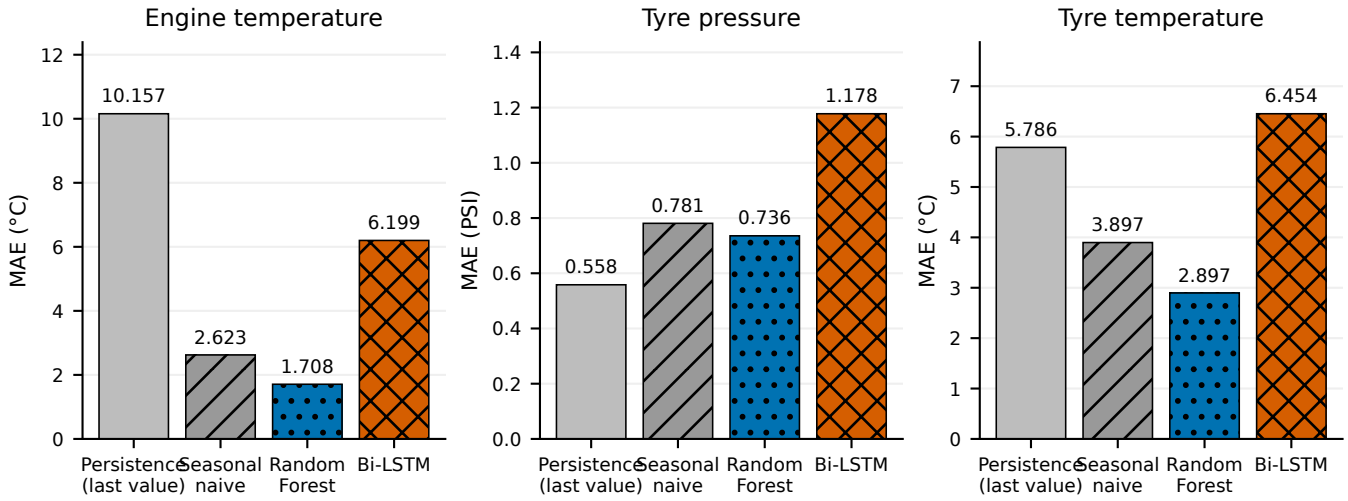



Fig. 2. Mean absolute error from the committed corrected results. Each panel has its own physical-unit axis. Hatching distinguishes models in grayscale.

window, take the median, and select the earliest window whose MAE is closest to that median. The selected zero-based test index is 1464, with first target at December 19, 2024, 00:00. Its Random Forest MAE is 1.579°C. This is representative only by that explicit criterion; it is not a substitute for the complete test table.

Fig. 4 resolves error by elapsed forecast time using aligned predictions. The engine has 24 hourly outputs, whereas each tyre target has eight outputs at three-hour intervals. The plotted values are computed from prediction errors at each horizon, not approximated from aggregate MAE. Persistence is included to put the learned methods in context. The differing curve shapes reinforce that a pooled score does not describe every lead time equally; no monotonic increase of error is assumed.

### C. Implications for Model Selection

Synthetic structure may favor simpler autoregressive or seasonal predictors, while the fixed network and limited training budget may not match each series well. These are possible explanations, not causal findings from an ablation study. The tyre-pressure epoch cap is a concrete reason not to interpret the ranking as proof that Bi-LSTM can never improve. Equally, its capacity does not justify selecting it without evidence.

For this benchmark, Random Forest is the empirical choice for the two temperature targets and persistence for pressure when MAE is the selection criterion. This retrospective ranking does not establish a production selection policy or an unbiased estimate of a newly tuned system. Deployment should evaluate model choices under a pre-declared validation procedure and verify cadence, sensor calibration, threshold meaning, and live data quality. The integrated prototype remains useful as a place to connect a suitable forecast model to a service process, irrespective of model family.

## VII. ROBUSTNESS AND ABLATION STUDIES

Table IV and Fig. 5 report all five predeclared Bi-LSTM initialization seeds rather than selecting a favorable seed. Mean

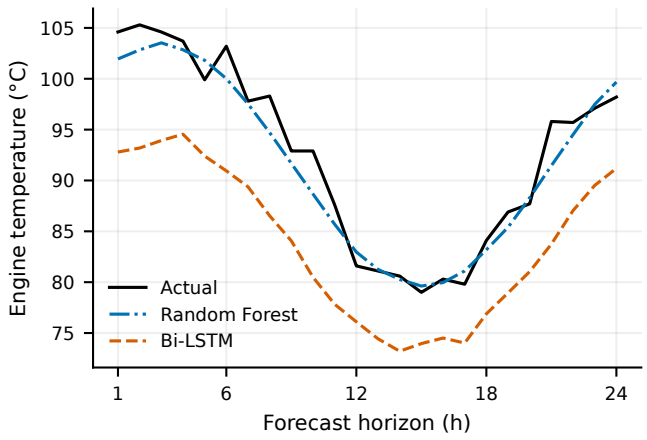



Fig. 3. Deterministically selected 24-hour engine forecast. Forecast issued at 2024-12-18 23:00:00 (dataset clock); selected RF window MAE 1.579 °C.

$R^2$ is negative for tyre temperature ($-0.835 \pm 0.536$), so the five-seed results do not support a blanket claim of Bi-LSTM superiority.

The seed-42 architecture ablation finds lower Bi-LSTM MAE for engine temperature (6.199 versus 9.539°C) and tyre pressure (1.182 versus 1.298 PSI), but lower LSTM MAE for tyre temperature (4.402 versus 6.366°C). Table V is therefore a controlled first-pass comparison, not a seed-robust architecture ranking. The causal feature ablation is shown in Table VI. F2 lowers engine-temperature MAE for both RF and LSTM relative to F0, whereas feature effects differ by target; no feature set was chosen from test results.

## VIII. Threshold-Event Evaluation and Uncertainty

Thresholds are derived from the AutoSense training region only. Table VII reports proxy-event scores, including the high tyre-temperature Bi-LSTM F1 (0.802) together with its false-positive rate of 1.0 and specificity of 0.0. It is consequently not a useful discriminative alert detector. These scores concern forecast excursions from data-derived proxy bounds, not component failures or certified engineering thresholds.

For the predeclared lowest-MAE corrected baseline against Bi-LSTM, the 2,000-replication non-circular moving-block bootstrap intervals for MAE(baseline) − MAE(Bi-LSTM) are $[-5.033, -3.955]$°Cfor engine temperature (RF), $[-0.845, -0.356]$ PSI for tyre pressure (persistence), and $[-3.732, -3.392]$°Cfor tyre temperature (RF). Every interval excludes zero, but dependence-aware intervals are not i.i.d. significance tests or evidence of operational benefit.

## IX. External Benchmark Evaluation on NASA C-MAPSS FD001

NASA C-MAPSS FD001 is an external *simulated* engine-degradation benchmark, not real measured flight telemetry. We forecast sensors 9, 11, 14, and 12 at horizons one through ten cycles, using the fixed disjoint 70/15/15 engine split. C0 uses target history only; C1 uses multivariate history. No engine occurs in more than one split, sensor selection uses training engines only, and the scaler is fitted on training engines only.

Table VIII gives training-range-normalized macro results, which avoid averaging incomparable native sensor units. In C1, Random Forest achieves the lowest macro normalized MAE (0.03406), lowest normalized RMSE (0.04278), and highest $R^2$ (0.88907) among the evaluated seed-42 model configurations. These values describe fixed configurations rather than a tuned model contest. The results were recorded with TensorFlow 2.20.0 and scikit-learn 1.6.1, unlike the 2.18.0/1.9.0 AutoSense runs.

The fixed early ($< 0.5$), mid ($0.5$ to $< 0.8$), and late ($\geq 0.8$) normalized-life bins are used only to stratify held-out sensor-forecasting error after prediction; they do not create an RUL target. Fig. 8 summarizes the C1 sensor, horizon, and stage views. The stage script itself records TensorFlow 2.18.0/scikit-learn 1.9.0, but its inputs are the accepted forecast artifacts recorded under TensorFlow 2.20.0/scikit-learn 1.6.1; that inherited environment deviation remains part of the provenance.

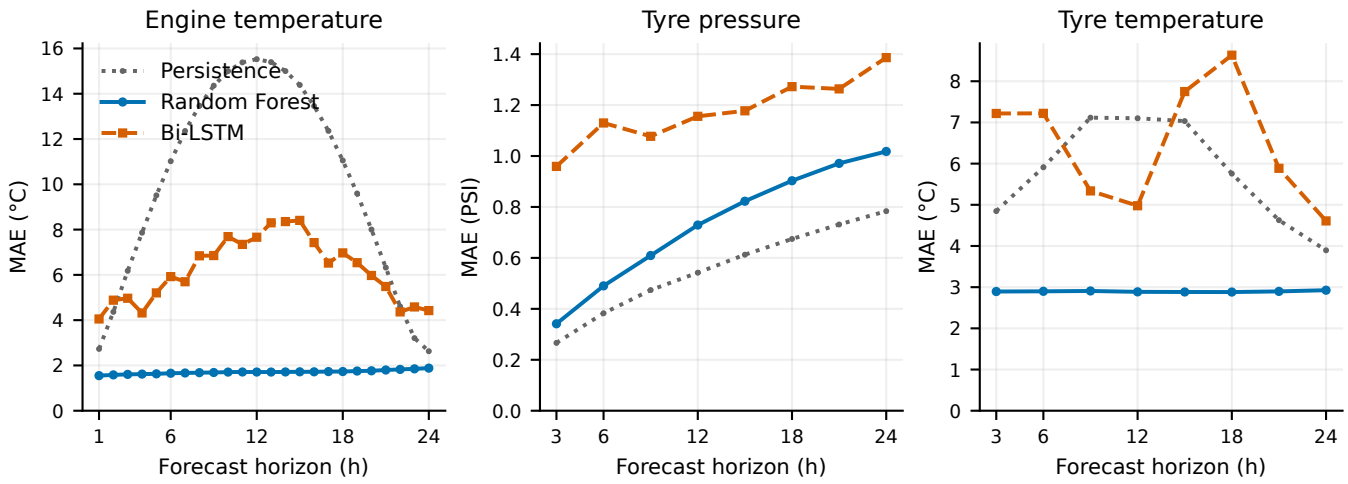



Fig. 4. Horizon-wise MAE for persistence, Random Forest, and Bi-LSTM. Engine horizons are 1–24 hours; tyre horizons are 3, 6, . . . , 24 hours. Each target uses its own physical-unit axis.

TABLE IV
FIVE-SEED AUTOSENSE BI-LSTM RESULTS. VALUES ARE MEAN ± SAMPLE STANDARD DEVIATION OVER SEEDS 11, 21, 42, 84, AND 168.

| Dataset | MAE | RMSE | $R^2$ | Unit |
|---|---|---|---|---|
| Engine temperature | 5.182 ± 0.665 | 6.343 ± 0.655 | 0.531 ± 0.099 | °C |
| Tyre pressure | 1.256 ± 0.221 | 1.532 ± 0.260 | 0.510 ± 0.164 | PSI |
| Tyre temperature | 5.731 ± 0.889 | 7.030 ± 1.082 | -0.835 ± 0.536 | °C |

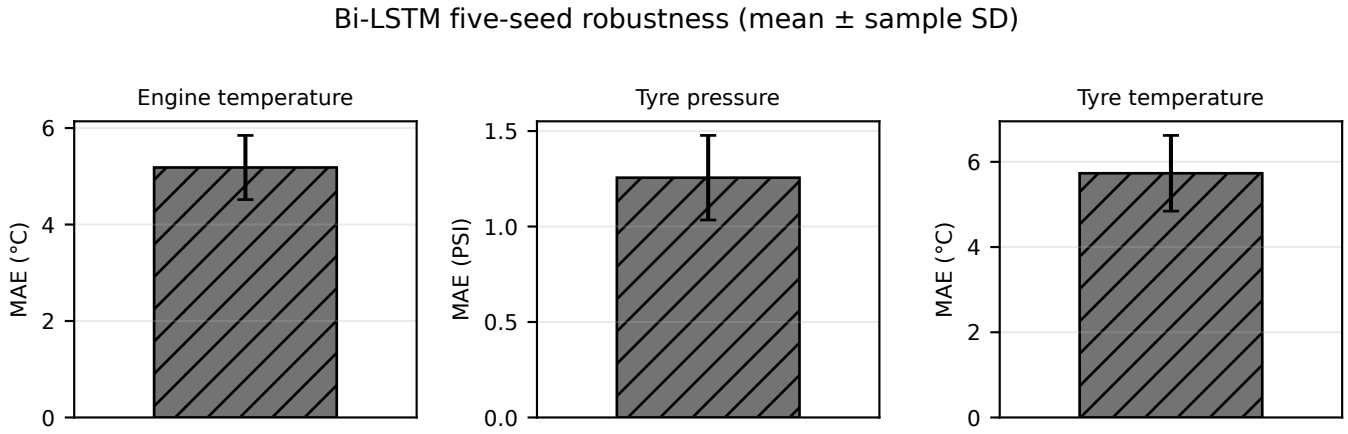



Fig. 5. Five-seed Bi-LSTM MAE. Each panel retains its native physical unit.

TABLE V
ARCHITECTURE ABLATION AT SEED 42; COMMON CHRONOLOGICAL SPLITS, TRAIN-ONLY SCALING, TRAINING BUDGET, AND MODEL DEFINITIONS ARE RETAINED.

| Dataset | LSTM MAE | Bi-LSTM MAE | LSTM $R^2$ | Bi-LSTM $R^2$ | Unit |
|---|---|---|---|---|---|
| Engine temperature | 9.539 | 6.199 | -0.604 | 0.374 | °C |
| Tyre pressure | 1.298 | 1.182 | 0.506 | 0.577 | PSI |
| Tyre temperature | 4.402 | 6.366 | -0.034 | -1.194 | °C |

TABLE VI
CAUSAL FEATURE-ABLATION MAE AT SEED 42. F0: TARGET HISTORY; F1: TARGET PLUS DETERMINISTIC CALENDAR; F2: F1 PLUS CAUSAL LAG/ROLLING FEATURES.

| Dataset | Model | F0 MAE | F1 MAE | F2 MAE | Unit |
|---|---|---|---|---|---|
| Engine temperature | RF | 1.709 | 1.710 | 1.673 | °C |
| Engine temperature | LSTM | 9.539 | 7.127 | 5.142 | °C |
| Tyre pressure | RF | 0.736 | 0.803 | 0.836 | PSI |
| Tyre pressure | LSTM | 1.212 | 2.401 | 2.279 | PSI |
| Tyre temperature | RF | 2.897 | 5.040 | 2.757 | °C |
| Tyre temperature | LSTM | 4.394 | 4.531 | 4.598 | °C |

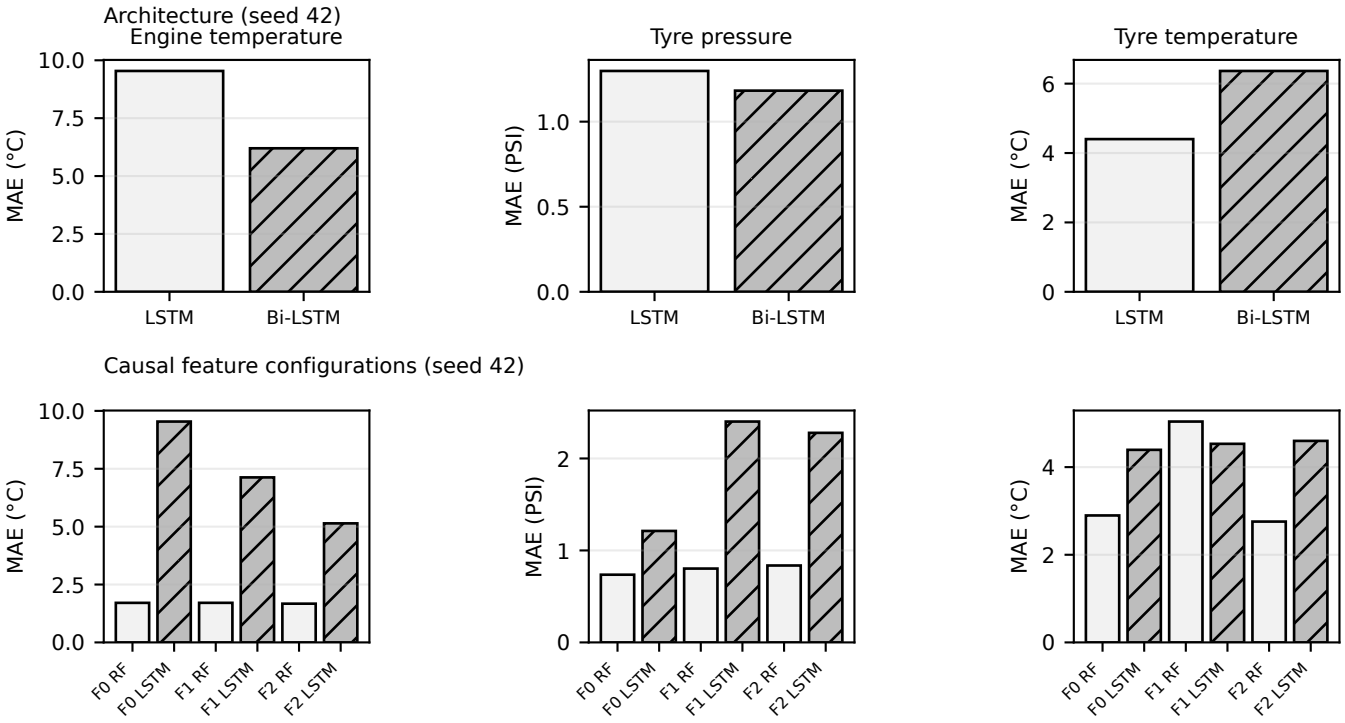



Fig. 6. Architecture and causal-feature ablations. Panels use separate native-unit axes; RF denotes Random Forest.

TABLE VII
TRAINING-DERIVED PROXY THRESHOLD-EVENT METRICS. FPR IS FALSE-POSITIVE RATE; THESE LABELS ARE NOT FAILURE LABELS.

| Dataset | Model | Precision | Recall | F1 | FPR |
|---|---|---|---|---|---|
| Engine temperature | Persistence | 0.429 | 0.016 | 0.031 | 0.003 |
| Engine temperature | Seasonal naive | 0.400 | 0.348 | 0.372 | 0.062 |
| Engine temperature | Random Forest | 0.000 | 0.000 | 0.000 | 0.000 |
| Engine temperature | Bi-LSTM | 0.000 | 0.000 | 0.000 | 0.000 |
| Tyre pressure | Persistence | 0.974 | 0.801 | 0.879 | 0.020 |
| Tyre pressure | Seasonal naive | 0.881 | 0.906 | 0.893 | 0.113 |
| Tyre pressure | Random Forest | 0.994 | 0.588 | 0.739 | 0.003 |
| Tyre pressure | Bi-LSTM | 0.000 | 0.000 | 0.000 | 0.000 |
| Tyre temperature | Persistence | 0.800 | 0.176 | 0.288 | 0.089 |
| Tyre temperature | Seasonal naive | 0.763 | 0.775 | 0.769 | 0.487 |
| Tyre temperature | Random Forest | 0.000 | 0.000 | 0.000 | 0.000 |
| Tyre temperature | Bi-LSTM | 0.670 | 1.000 | 0.802 | 1.000 |

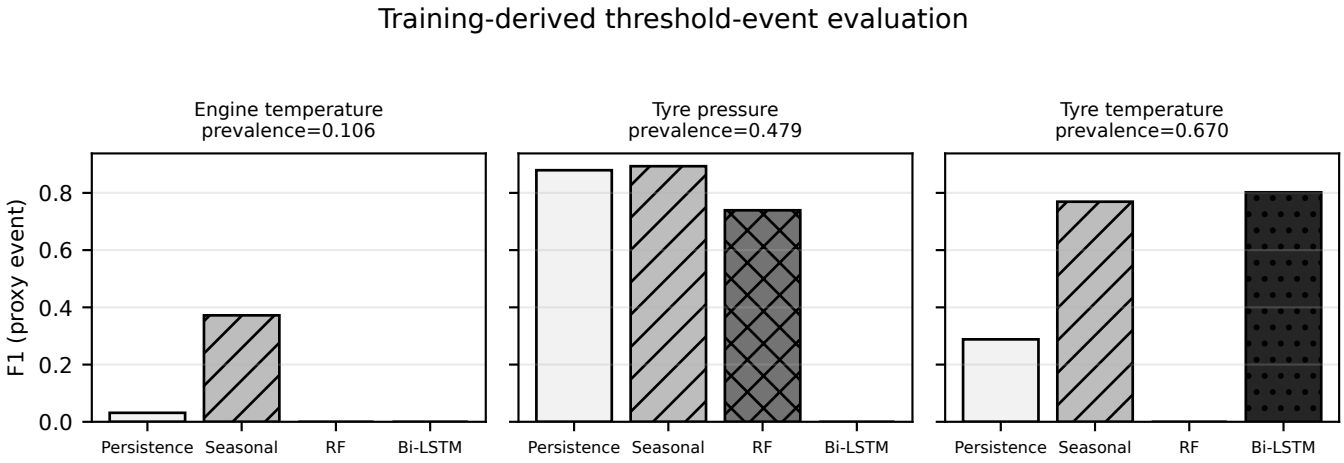



Fig. 7. Proxy-event F1 with the event prevalence shown per target.

TABLE VIII
C-MAPSS FD001 MACRO-AVERAGE TRAINING-RANGE-NORMALIZED METRICS. BI-LSTM USES THE THREE PREDECLARED SEEDS; C0 IS TARGET-ONLY AND C1 IS MULTIVARIATE HISTORY.

| Configuration | Model | Seed | NMAE | NRMSE | $R^2$ |
|---|---|---|---|---|---|
| C0 | Bi-LSTM | 11 | 0.03543 | 0.04442 | 0.87980 |
| C0 | Bi-LSTM | 42 | 0.03574 | 0.04481 | 0.87820 |
| C0 | Bi-LSTM | 84 | 0.03654 | 0.04582 | 0.87163 |
| C0 | LSTM | 42 | 0.03582 | 0.04490 | 0.87747 |
| C0 | Persistence | 42 | 0.04722 | 0.05988 | 0.78467 |
| C0 | Random Forest | 42 | 0.03472 | 0.04362 | 0.88438 |
| C1 | Bi-LSTM | 11 | 0.03499 | 0.04387 | 0.88457 |
| C1 | Bi-LSTM | 42 | 0.03483 | 0.04366 | 0.88536 |
| C1 | Bi-LSTM | 84 | 0.03505 | 0.04393 | 0.88414 |
| C1 | LSTM | 42 | 0.03460 | 0.04341 | 0.88620 |
| C1 | Random Forest | 42 | 0.03406 | 0.04278 | 0.88907 |

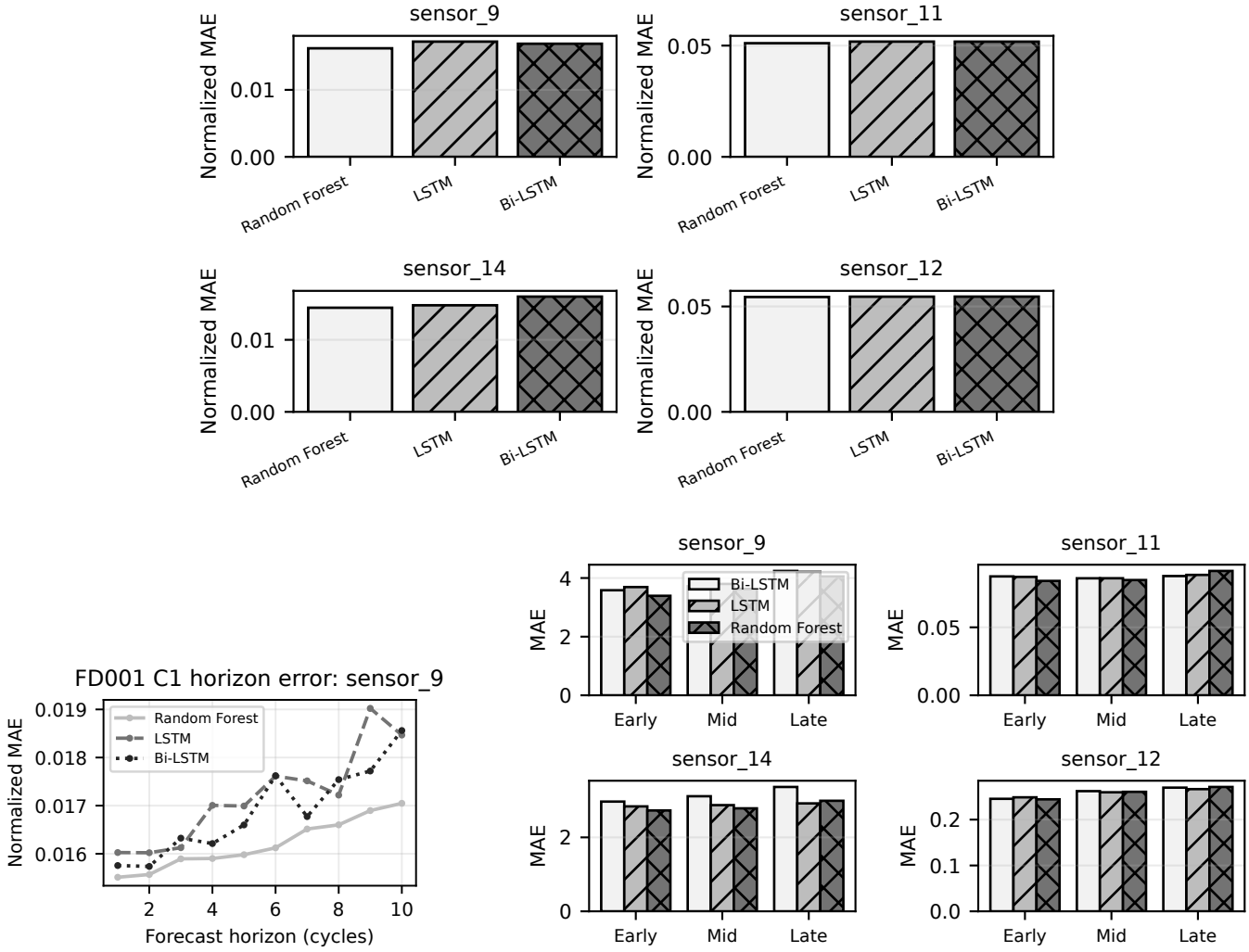



Fig. 8. C-MAPSS FD001 C1 results: top, normalized MAE by target sensor; the displayed Bi-LSTM bar is seed 42, while all three predeclared seeds remain in Table VIII. Bottom left, normalized MAE by forecast horizon for sensor 9 (Bi-LSTM seed 42). Bottom right, MAE stratified by fixed normalized-life stage; panels use native units and display Bi-LSTM seed 42 for readability.

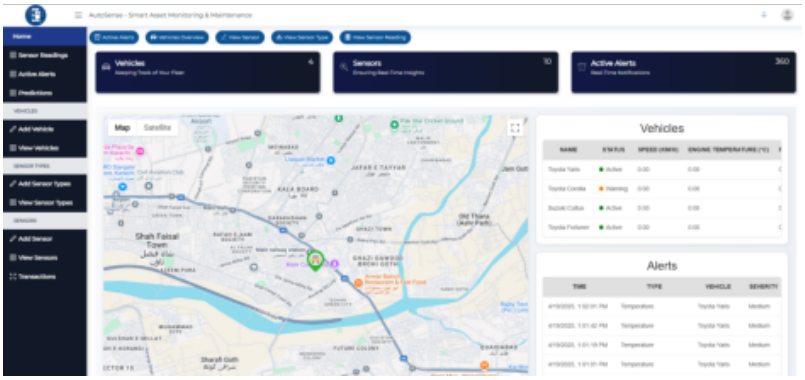
## X. Prototype Integration and Workflow Validation

The repository provides three complementary forms of integration evidence. First, firmware implements sensor parsing, telemetry payloads, and platform transmission. Second, platform scripts and interface captures demonstrate the intended ingestion and forecast-display interfaces. Third, Salesforce Apex implements alert import and vehicle-status callbacks. These artifacts support technical feasibility at the prototype level; the present work inspects them without contacting live endpoints or rerunning hardware and Salesforce tests.

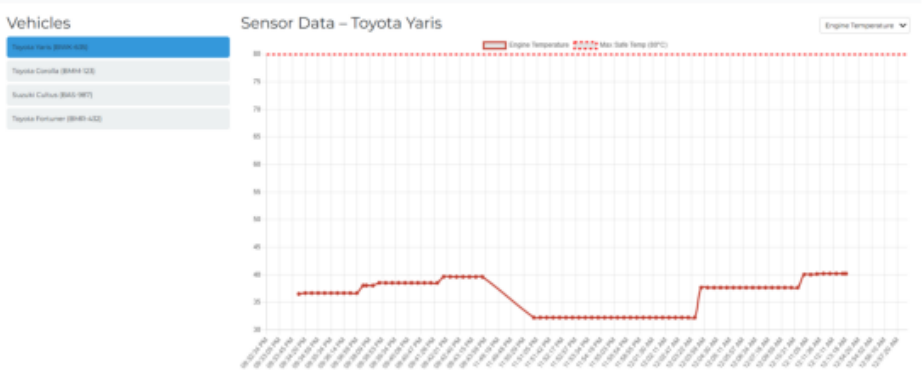
Fig. 9 retains the original fleet-map, sensor-view, and Salesforce Vehicle Alert screenshots. They establish the existence and layout of monitoring interfaces. Displayed vehicle names, readings, and thresholds are demonstration context, not the held-out synthetic benchmark or a validated operating fleet. The forecast-display screenshot is omitted to avoid confusing legacy forecast curves with the corrected comparison.

The alert-ingestion service maps identifiers, severity, description, and status to a custom Vehicle Alert record. The separate Salesforce panel in Fig. 9 shows that record type; it is not a screenshot of a Salesforce Case. A separate trigger detects changes to a vehicle's status and calls the status service. This demonstrates the implemented feedback mechanism in source code, but not that closing a Case caused that update in a reproducible test.

(a) Fleet map and monitoring interface




(b) Sensor view with configured threshold



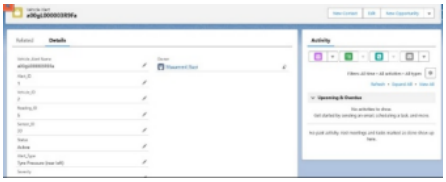



Fig. 9. Retained prototype interface evidence from the original project: (a) fleet map and monitoring, (b) sensor view with a configured threshold, and (c) a Salesforce Vehicle Alert record. The screenshots are interface demonstrations, not scientific test observations, verified Case creation, or verified Flow automation. The source raster for (c) is low resolution.

Case creation and assignment through Flow, the resolution-to-vehicle linkage, and live forecast-endpoint operation are described in the original project but lack sufficient exported configuration or execution traces for independent reproduction. No latency distribution, success-rate denominator, environmental-test record, or measured downtime/cost study was found in the reviewed evidence. Consequently, the paper makes no numerical integration-reliability or response-time claim. A future integration test should trace a timestamped sensor event through alert creation, Case assignment, closure, and dashboard feedback with recorded outcomes.

## XI. Limitations

The principal limitation is synthetic data: the results measure forecasting on generator-produced signals rather than external real-fleet observations. The hardware and software artifacts do not establish domain transfer, automotive robustness, or aviation operational validation. Sensor placement and calibration also determine whether a displayed temperature is a useful proxy for engine condition.

The corrected benchmark's main comparison uses one chronological holdout and a fixed limited training budget with no full hyperparameter search. Five-seed variation is reported only for the AutoSense Bi-LSTM, and the LSTM/Bi-LSTM architecture comparison remains a seed-42 first pass. Overlapping forecast origins yield dependent errors; the moving-block intervals address this dependence for the predeclared baseline-versus-Bi-LSTM MAE comparisons but do not convert the study into repeated-split validation. Negative $R^2$ and physical-unit errors are retained rather than replaced by normalized metrics.

The C-MAPSS study is an external simulated benchmark rather than a real fleet or measured flight-telemetry validation. It uses a different recorded TensorFlow/scikit-learn environment from some AutoSense phases; version heterogeneity

limits exact cross-environment reproducibility even though the pre-registered sources, splits, sensors, scalers, settings, hashes, and prediction-engine IDs were validated. Feature-ablation conclusions likewise belong to their recorded TensorFlow 2.20.0/scikit-learn 1.6.1 environment.

There are no maintenance-outcome labels, direct failure or remaining-useful-life targets, or measured downtime and cost reductions. Forecast accuracy alone cannot establish operational maintenance benefit or calibrated alarm risk. Case/Flow configuration and live transaction logs are incomplete in the repository, so the fully closed service loop remains less strongly evidenced than its constituent ingestion and status components. The legacy integration scripts have not been migrated to the corrected benchmark protocol.

## XII. Conclusion and Future Work

This work presents an integrated IoT-to-enterprise prototype and a corrected comparative study of 24-hour multi-step sensor forecasting. The available firmware, platform interfaces, and Salesforce code support prototype-level technical feasibility, with explicit limits on workflow validation. On three synthetic datasets, Random Forest provides the lowest temperature MAEs and persistence the lowest pressure MAE. Five-seed robustness, fixed architecture and causal-feature ablations, proxy threshold-event metrics, and moving-block intervals reinforce the need for evidence-based model selection. A separate, engine-disjoint C-MAPSS FD001 simulated-benchmark study extends the analysis to multivariate sensor forecasting, while retaining its environment deviation.

Future work should prioritize real fleet data collection, genuine failure and maintenance labels, and external validation. Uncertainty-aware forecasts and richer CAN bus features should be assessed under a pre-declared model-selection protocol with strong baselines. More complex architectures, including Transformers, would require that same evidential standard. Completing and exporting Case/Flow configuration, validating the live feedback loop, and measuring maintenance outcomes would connect forecast quality to practical service value.

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